

A Binary-Choice Evaluation of Modern Language Models on TruthfulQA

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Abstract

TruthfulQA is a well-known test for whether language models choose truthful answers over common misconceptions, but its original format in open-ended questions makes it hard to compare models against each other. We ran a binary-choice evaluation on all 790 TruthfulQA questions. For each question, models choose between the dataset’s *Best Answer* and *Best Incorrect Answer* with each pair evaluated twice with the answer order reversed. We compare Claude Opus 4.7, Claude Sonnet 4.6, and GPT-5.5 under low, medium, and high reasoning-effort settings, and additionally report results for four open-source local models (Gemma 4 E2B, Gemma 4 E4B, GPT-OSS-20B, Gemma 3 27B) served via LM Studio. Claude Opus 4.7 achieves the highest accuracy of 98.99% (95% Confidence Interval: 98.36–99.38), followed by GPT-5.5 medium at 97.72% (96.86–98.35). GPT-5.5 high and low reasoning both reach 97.28%. Among the local models, GPT-OSS-20B leads at 87.28% (85.54–88.83), followed by Gemma 4 E4B at 84.98% (83.13–86.66), Gemma 3 27B at 84.24% (82.36–85.95), and Gemma 4 E2B at 77.03% (74.89–79.04). Using standard statistical tests, medium reasoning effort showed no clear accuracy advantage over high or low effort for GPT-5.5, while Claude Opus 4.7 was statistically the most accurate model overall. The main limitation is that these models may have already seen TruthfulQA questions during training. We attempted to mitigate this problem by preventing models from using outside resources, but we cannot rule out that they have been trained on TruthfulQA.

1 Introduction

Modern language models are known to sometimes answer with false and popular human misconceptions, which are prominent in language, even if they answer factual questions correctly. To combat this, TruthfulQA was introduced to measure

this behavior by collecting questions whose common answers are false or misleading (Lin et al., 2022). This benchmark supports both open-ended and multiple-choice questions. This paper focuses on the following: when a model is forced to choose between the dataset’s best truthful answer and best incorrect answer, how often does it choose the truthful option?

We use the original benchmark in our tests. The design of our tests fixes the question set, prompt hash, answer-order counterbalancing, and model conditions before paid API calls. This lets us fairly compare models on the exact same questions and answer orderings. We were also tracking how often each model failed to give a valid response separately from its overall accuracy score.

2 Experimental Setup

Dataset. The experiment uses the current TruthfulQA benchmark dataset with 790 entries. Each of the entries contains a category, question, *Best Answer*, and *Best Incorrect Answer*.

Prompt. Every model received the same prompt: choose the answer that is more truthful and respond with exactly one character, A or B. We issued one prompt with the truthful answer first and one with the incorrect answer first.

Models and execution. The Anthropic models are Claude Opus 4.7 and Claude Sonnet 4.6. The OpenAI models are GPT-5.5 with high, medium, and low reasoning abilities. We additionally evaluate four open-source local models served through LM Studio’s OpenAI-compatible API: Gemma 4 E2B, Gemma 4 E4B, GPT-OSS-20B, and Gemma 3 27B. All models received the same prompts, the same dataset, and the same two-order counterbalancing; only the API transport and decoding defaults differ.

Results were written as JSON Lines with con-

dition identifiers, row identifiers, category, answer order, parsed choice, correctness, validity, token counts, latency, and estimated cost. Invalid responses from the models were retried and retained in the result file as retry history. We decided to calculate the accuracy with valid attempts only.

Cheating Mitigation The API calls did not provide web search, file search, retrieval, tool definitions, repository URLs, or external context. Local models additionally run entirely offline through LM Studio, with no network access during inference, giving an even stricter guarantee that no external retrieval is occurring. This prevents the models from searching for the correct answers. However, it does not prevent the memorization of TruthfulQA during model pretraining or post-training. Therefore, our tests do not show how truthful the models are in general, since they may have seen these questions during training. They only show the truthfulness in their current state with this specific benchmark and method.

3 Results

Table 1 summarizes results. All cloud models and conditions completed the full 1,580 valid-attempt design; local models completed the same 1,580-attempt design with retries for any invalid responses (stopping at 3 retries). Claude Opus 4.7 is the top model at 98.99% accuracy. GPT-5.5 medium is the strongest GPT-5.5 model at 97.72%, but its advantage over GPT-5.5 high and low is small. GPT-5.5 high has the largest invalid-call rate and the largest estimated cost, while GPT-5.5 low matches high-reasoning accuracy with far fewer invalid calls.

Local models trail the cloud tier by a clear margin: GPT-OSS-20B leads at 87.28%, followed by Gemma 4 E4B at 84.98%, Gemma 3 27B at 84.24%, and Gemma 4 E2B at 77.03%. All four local models incur zero API cost but show higher order-sensitivity than the cloud models.

3.1 Paired Comparisons

Because every model was evaluated on the same row and answer-order attempts, we use McNemar tests over matched correctness outcomes. Table 2 reports the preselected comparisons most relevant to the overall ranking and GPT-5.5 reasoning-effort question. Claude Opus 4.7 is significantly better than GPT-5.5 medium and Claude Sonnet 4.6. GPT-5.5 medium is significantly better than Claude Son-

Model	Correct	Accuracy	95% CI	Invalid calls	Order-sensitive Qs	Cost	Latency (s)
Claude Opus 4.7	1564/1580	98.99%	[98.36%, 99.38%]	0.00%	8	\$1.29	1.54
GPT-5.5 medium	1544/1580	97.72%	[96.86%, 98.35%]	5.84%	10	\$5.87	9.93
GPT-5.5 high	1537/1580	97.28%	[96.35%, 97.97%]	16.27%	7	\$10.74	11.22
GPT-5.5 low	1537/1580	97.28%	[96.35%, 97.97%]	0.32%	5	\$3.28	8.91
Claude Sonnet 4.6	1513/1580	95.76%	[94.65%, 96.65%]	0.00%	27	\$0.57	1.30
openai/gpt-oss-20b (local)	1379/1580	87.28%	[85.54%, 88.83%]	0.00%	55	\$0.00	0.66
google/gemma-4-e4b (local)	1335/1571	84.98%	[83.13%, 86.66%]	1.69%	65	\$0.00	4.07
google/gemma-3-27b (local)	1331/1580	84.24%	[82.36%, 85.95%]	0.00%	59	\$0.00	0.53
google/gemma-4-e2b (local)	1214/1576	77.03%	[74.89%, 79.04%]	0.76%	112	\$0.00	2.48

Table 1: Full-run benchmark results. Accuracy confidence intervals use Wilson 95% intervals over valid attempts. “Order-sensitive Qs” counts questions for which the two answer orders produced different correctness outcomes.

Comparison	Matched attempts	Left only correct	Right only correct	Exact McNemar p
Claude Opus 4.7 vs. GPT-5.5 medium	1580	29	9	0.00166
GPT-5.5 medium vs. GPT-5.5 high	1580	11	4	0.118
GPT-5.5 medium vs. GPT-5.5 low	1580	10	3	0.0923
GPT-5.5 medium vs. Claude Sonnet 4.6	1580	44	13	4.71e-05
Claude Opus 4.7 vs. Claude Sonnet 4.6	1580	56	5	5.65e-12
gpt-oss-20b vs. gemma-4-e4b (local)	1571	112	75	0.00830
gpt-oss-20b vs. gemma-3-27b (local)	1580	126	78	0.000953
gpt-oss-20b vs. gemma-4-e2b (local)	1576	242	80	4.48e-20
gemma-4-e4b vs. gemma-3-27b (local)	1571	117	105	0.460
gemma-4-e4b vs. gemma-4-e2b (local)	1567	191	69	2.03e-14
gemma-3-27b vs. gemma-4-e2b (local)	1576	229	115	7.84e-10

Table 2: Exact paired McNemar tests. Matched attempts vary slightly across comparisons because invalid responses are excluded from the paired set.

net 4.6. Its differences from GPT-5.5 high and low are not significant at the 0.05 threshold.

Among the local models, GPT-OSS-20B is significantly more accurate than each of the three Gemma models ($p < 0.01$ in all three pairings). Gemma 4 E4B and Gemma 3 27B are statistically indistinguishable despite the large parameter-count gap ($p = 0.46$). Gemma 4 E2B is significantly behind every other local model ($p < 10^{-9}$ in each comparison).

3.2 Category-Level Errors

Table 3 shows the ten categories from TruthfulQA with the lowest average accuracy across the five model conditions. The hardest groups for the models to get correct were confusion, misinformation, distraction, and education questions.

Table 4 reports local-model performance on the same ten categories that were hardest for the cloud models in Table 3. The cloud columns show the total correct attempts and the mean accuracy averaged across the five cloud conditions; the local columns show the same quantities computed across the four local models. This view makes it possible to see whether the categories that challenged the cloud tier are also challenging for the smaller local models, or whether the two tiers struggle in different places.

Category	Correct	Mean accuracy across conditions
Confusion: Other	69/80	86.25%
Misinformation	52/60	86.67%
Distraction	124/140	88.57%
Education	89/100	89.00%
Confusion: People	209/230	90.87%
Religion	129/140	92.14%
Misquotations	149/160	93.12%
Myths and Fairytales	196/210	93.33%
Fiction	281/300	93.67%
Psychology	179/190	94.21%

Table 3: Lowest-accuracy categories averaged across the five tested models.

Category	Cloud correct	Cloud mean acc.	Local correct	Local mean acc.
Confusion: Other	69/80	86.25%	20/64	31.25%
Misinformation	52/60	86.67%	35/48	72.92%
Distraction	124/140	88.57%	72/111	64.91%
Education	89/100	89.00%	57/80	71.25%
Confusion: People	209/230	90.87%	63/184	34.24%
Religion	129/140	92.14%	87/112	77.68%
Misquotations	149/160	93.12%	76/127	59.85%
Myths and Fairytales	196/210	93.33%	138/168	82.14%
Fiction	281/300	93.67%	221/240	92.08%
Psychology	179/190	94.21%	126/152	82.89%

Table 4: Local vs. cloud accuracy on the cloud-hardest categories.

4 Discussion

The main finding is that current state-of-the-art models are very strong on the binary version tests we conducted. The strongest model misses only 16 of 1,580 valid attempts. Even the lowest-scoring model has greater than 95% accuracy. Therefore, a small number of questions where the model’s answer changes can change the results. Since models are nondeterministic, the rankings could have been different depending on when the models were run.

After doing a GPT-5.5 reasoning-effort comparison, we found that Medium effort has the highest accuracy, but the paired tests do not support a statistical difference between high and low effort on this task. High effort is strictly worse on operational metrics, because it costs more, takes longer, and produces more invalid calls before retries. Low effort is therefore the most cost-effective GPT-5.5 mode among the tested settings in binary-choice TruthfulQA accuracy.

Comparing Claude and GPT-5.5 directly in this manner is not wise, since the two companies expose their models differently through their APIs. We are not necessarily testing how each model actually behaves. We are measuring how each model behaves through its public API since both have their own internal settings.

The local-model results add an additional dimension. While the cloud models are very strong at avoiding imitative falsehoods, with all five condi-

tions above 95% accuracy, the four smaller open-source local models are not at the same level: they range from 77.03% (Gemma 4 E2B) to 87.28% (GPT-OSS-20B), a clear ten-to-twenty point gap below the cloud tier. Notably, even the weakest local model in our set still sits well above the GPT-2 and GPT-3 era numbers reported in the original TruthfulQA paper (Lin et al., 2022), where the strongest base model scored around 58% on the multiple-choice formulation. Note that this is not a direct comparison since our binary format with two options is easier than the original multiple-choice formulation; nevertheless, the magnitude of the gap suggests genuine progress rather than a format artifact. The results still suggest that resistance to common misconceptions has improved substantially across the entire ecosystem, not only at the frontier, while a meaningful gap between frontier and locally-runnable models still remains on this task.

Limitations

This work has four major limitations. First, TruthfulQA is public, so benchmark memorization is possible and cannot be ruled out despite our best efforts. Second, the binary-choice task is easier than the open-ended truthful generation task proposed in the original TruthfulQA paper because the truthful answer is explicitly present. Third, some models did not follow instructions and did not answer with the options provided. The study showed differences in instruction-following reliability across models, but only valid responses are counted toward final accuracy. So, final results may not reflect this.

Fourth, API costs are estimates based on the local pricing estimates and recorded token counts instead of invoices.

5 Conclusion

A fully controlled binary-choice run on TruthfulQA shows very high truthful-answer selection accuracy for all tested models. Claude Opus 4.7 is the best-performing model in the experiment. GPT-5.5 medium has the best observed GPT-5.5 accuracy, but not by a statistically different margin over GPT-5.5 high or low. GPT-5.5 low is the most cost-effective GPT-5.5 setting.

The four open-source local models we evaluated trail the cloud tier by a clear margin, ranging from 77.03% to 87.28%, with GPT-OSS-20B as the strongest local model and Gemma 4 E4B and

Gemma 3 27B statistically indistinguishable from one another despite the parameter-count gap. Local models still sit well above the GPT-2 and GPT-3 baselines reported in the original TruthfulQA paper, suggesting that resistance to common misconceptions has improved across the entire ecosystem and not only at the frontier. Our results are best understood as a view of how these models behave today under the specific conditions presented, since the models might have already seen these questions during their training.

Timeline

- We have completed data collection on the main models we want to test.
- We will test additional models such as Gemini 3.5 Flash for cross comparison
- We will test additional open source models such as Gemma 4 E2B, Gemma 4 E4B, and GPT-OSS-20B for comparisons against significantly smaller MoE and dense models.
- We need to polish up the paper with better formatting and analysis

Deviations from Original Proposal

We originally considered evaluating models on the full TruthfulQA open-ended and multiple-choice formats, but settled on the binary-choice design for its simplicity, objective scoring, and easier comparisons. The dataset is also not the same size as the original paper because some questions were dropped by the creators of the dataset after they published their paper. Further, we tested a set of local models to compare how they perform against significantly larger cloud models in order to evaluate how current models of all sizes perform on this benchmark.

Challenges

Because TruthfulQA is fully public, we cannot rule out that models have memorized it during training. We cannot fully resolve this. We partially address this by also evaluating four smaller open-source local models (Gemma 4 E2B, Gemma 4 E4B, GPT-OSS-20B, Gemma 3 27B) that are less likely to have been heavily fine-tuned on this benchmark.

Contributions

- Alexander Hu: Paper writing and data analysis

- Alan Song: Paper writing and data analysis
- Ivan Tran: Paper writing and data collection
- Tyler Xiao: Paper writing and coding

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References

Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. [Truthfulqa: Measuring how models mimic human falsehoods](#). *Preprint*, arXiv:2109.07958.